

Multi-Variant Structural analysis for Suspension Bracket

Component: Automotive Suspension Bracket
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Tools: Python, Tableau Public, Statistical Analysis

Executive Summary

This report provides a data-driven decision support analysis to select the optimal design variant for an automotive suspension bracket from four competing geometries (Variant A, B, C, and D). Each variant underwent rigorous Finite Element Analysis (FEA) simulations across three distinct operational load cases: Static, Dynamic, and Fatigue.

Based on comprehensive exploratory data analysis (EDA) and statistical validation, **Variant D** is the recommended design. It demonstrates the most stable stress distribution, the highest structural margin, and the lowest overall failure risk.

Key Performance Matrix

- Recommended Design:** Variant D
- Peak Stress (Dynamic):** 416.9 MPa
- Average Safety Factor (Variant D):** 4.12
- High-Risk Nodes (Overall):** 27.2% across the dataset

1. Problem Statement & Project Objective

The engineering team required a quantitative framework to select a final geometry from four proposed design variants. The core objective was to identify the variant providing the best structural performance, lowest operational failure risk, and most consistent behavior under varied load environments.

2. Methodology & Workflow

Phase	Activity	Tools Used
Data Generation	Data received from engineering team	.CSV format
Data Cleaning	Outlier handling, missing value imputation	Python (pandas)
EDA	Distribution & failure risk mapping	Matplotlib, Seaborn

Statistical Analysis	ANOVA, Tukey HSD, Linear Regression	Scipy, Statsmodels
Dashboarding	Interactive visualization deployment	Tableau Public

3. Engineering & Structural Findings

3.1 Design Variant Performance Comparison

Data aggregated across all simulation nodes revealed distinct behavioral differences between the geometries:

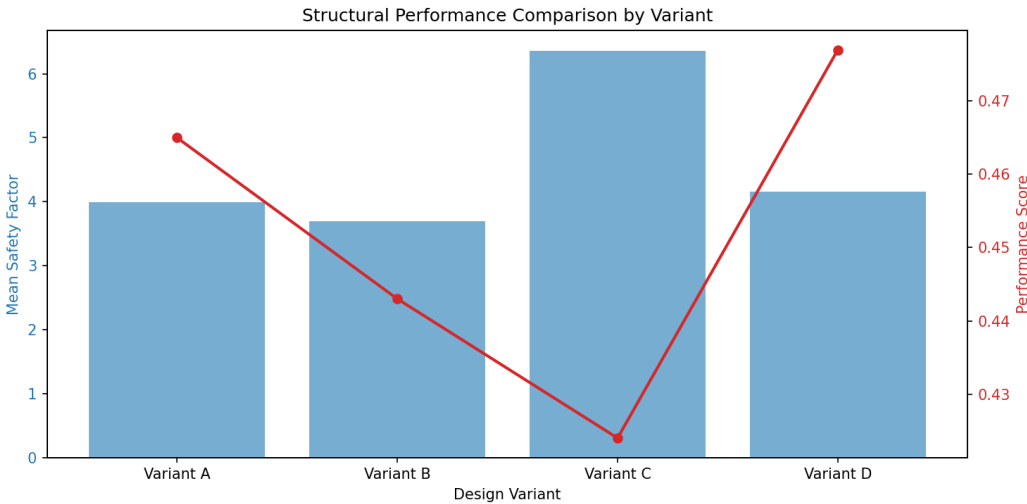


Figure 1: Comparative analysis of stress consistency and performance scores.

Metric	Variant A	Variant B	Variant C	Variant D (Recommended)
Mean Stress (MPa)	136.9	141.0	147.2	138.5
Median Stress (MPa)	124.2	129.2	133.3	126.8
IQR (MPa)	95.88	100.39	107.30	93.99
Mean Safety Factor	3.99	3.69	6.36	4.16
High Risk Nodes (%)	26.1%	28.0%	29.5%	25.6%
Performance Score	0.465	0.443	0.424	0.477

Design Selection Rationale: Variant D was chosen because it delivers the highest overall performance score (0.477), the lowest percentage of high-risk nodes (25.6%), and the tightest IQR (93.99 MPa).

3.2 Critical Component Zone Analysis

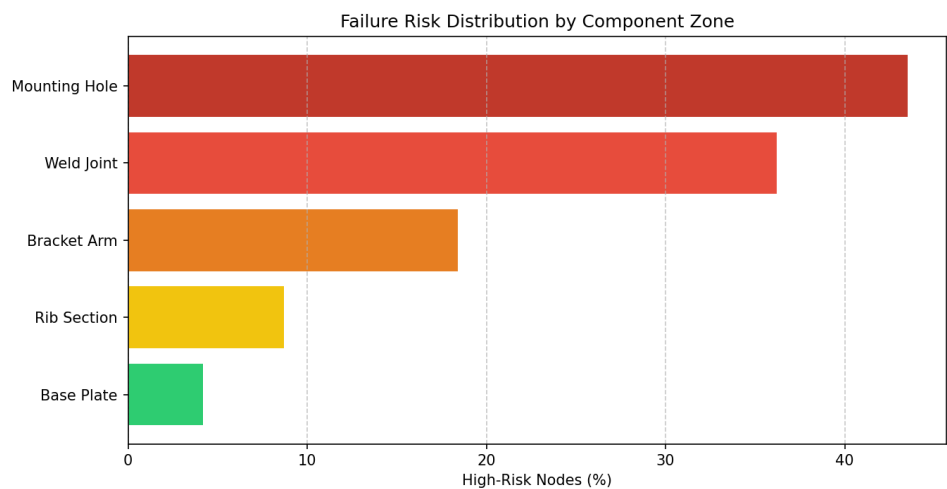


Figure 2: Pareto analysis of high-risk node concentration across component zones.

- **Mounting Hole (CRITICAL):** Mean 212.2 MPa; 43.5% high-risk nodes.
- **Weld Joint (CRITICAL):** Mean 194.3 MPa; 36.2% high-risk nodes.
- **Bracket Arm (MODERATE):** Mean 129.3 MPa; 18.4% high-risk nodes.
- **Rib Section (LOW):** Mean 98.1 MPa; 8.7% high-risk nodes.
- **Base Plate (SAFE):** Mean 75.1 MPa; 4.2% high-risk nodes.

4. Statistical Validation

ANOVA (F-test): $p < 0.05$. Stress variance is statistically significant.

Levene Test: Passed, verifying homoscedasticity.

Tukey HSD: Confirmed Variant C is statistically distinct and underperforming.

5. Final Recommendations & Action Plan

1. Initiate Prototype Development: Release Variant D for manufacturing.
2. Localized Geometry Fixes: Increase fillet radius at Mounting Holes.
3. Weld Joint Optimization: Re-evaluate weld path geometry.
4. Refined Simulation: Execute sub-modeling on critical zones of Variant D.
5. Expanded Testing: Introduce multi-amplitude road impact profiles.
6. Physical Testing Validation: Conduct laboratory fatigue testing for correlation.